POSTGRESQL · ASSIGNMENT 2

Group Food Order

Grade 11 · due in one week, before the next lesson · handed in on GitHub, folder **assignment-2**

Online version: tinyurl.com/tues-hw-2 database: food

1. The task

The class often orders food together – from the canteen or a pizza place. There is a menu with prices, everyone says what they want and how many, someone collects the money, and in the end there is always an argument about who owes how much. Build a database that remembers **the menu**, **every group order** and **who ordered what in it**, so that the bill comes out of the database and not out of a notebook.

One order contains many items from many people. One person can order several different things and several pieces of the same thing.

The database must be able to answer

1. For a given order: who owes how much? (name, amount)
2. The total of every order, sorted by date.

Each question = one SQL query that returns the answer as a table.

Minimum data

- at least 6 menu items with prices (with a decimal part!)
- at least 5 students
- at least 2 orders on different dates
- in one of the orders at least one person ordered 2+ pieces of something and at least one person ordered two different things

Your made-up data is good when every query returns at least 2 rows and there is something that could go wrong.

2. What you hand in

In your GitHub repository `tues-postgres` (one for the whole year) create a folder `assignment-2` with three files:

FILE	WHAT IT CONTAINS
`README.md`	3–5 sentences, "Decisions I made and why": which tables, which keys, what I deliberately do not store. In your own words.
`dump.sql`	A full export of the `food` database made with `pg_dump` (structure + data). I restore it on my machine and check it.
`queries.sql`	The queries for the questions in section 1, with a comment above each one saying which question it answers and what it returns.

AI is allowed. One condition: you show the schema and explain the solution yourself. If you cannot say why a table or a column exists, the homework is not yours.

What I check: the dump restores without errors · every table has a PRIMARY KEY · relationships use FOREIGN KEY · required columns are NOT NULL · the types fit (dates are DATE, money is NUMERIC) · both queries return the right answer · the README explains why.

3. How to submit

1. In pgAdmin create the database `food`; inside it make the tables, insert the data and write the queries (Query Tool).
2. Export the whole database (structure + data) to `dump.sql`. In a terminal, with Docker running:

   ```
   docker exec local-postgres pg_dump -U postgres -d food -f /tmp/dump.sql
   docker cp local-postgres:/tmp/dump.sql dump.sql
   ```

   The file appears in the folder where you ran the commands.
3. Put everything on GitHub: repository `tues-postgres`, folder `assignment-2`, the three files. Open the folder in your browser and check that all three are there.

Step-by-step GitHub instructions (account, repository, upload): tinyurl.com/tues-hw-guide

Before you submit

- [ ] Every table has `id … PRIMARY KEY` or a composite PRIMARY KEY.
- [ ] Every column that points to another table has `REFERENCES`.
- [ ] I tried to insert a row with a wrong id to another table and the database refused.
- [ ] Both queries return what I expect with my made-up data.
- [ ] The README answers "why", not only "what".